5-MINUTE REVISION

Sample Survey

From the chapter's own revision points, formulae and questions.

Selected from Chapter 12 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Sample survey → systematic collection of information from a sample.
- Population → complete collection of units under study.
- Population size → N.
- Sample → subset of the population selected for study.
- Sample size → n.

Quick revision · 2

- Sampling fraction → $\frac{n}{N}$.
- Sampling unit → basic unit selected from the population.
- Census → complete enumeration of every population unit.
- Sampling design → procedure used to select the sample.
- Sampling frame → complete list or source identifying the sampling units.

Quick revision · 3

- Parameter → numerical characteristic of the population.
- Statistic → numerical measure calculated from sample data.
- Estimator → statistic used to estimate a population parameter.
- Estimate → particular value obtained from an estimator.
- Unbiased estimator → estimator whose expected value equals the true parameter.

Quick revision · 4

- Sampling error → difference caused by observing a sample rather than the entire population.
- Sampling error is absent in a census.
- Increasing sample size generally decreases standard error for a fixed design and population.
- Non-sampling error → errors other than sampling error.
- Census data can still contain non-sampling errors.

Quick revision · 5

- Confidence interval → interval estimate of a population parameter.
- Confidence level → $100(1-\alpha)\%$.
- $\alpha = 0.05$ → 95% confidence level.
- $\alpha = 0.01$ → 99% confidence level.
- A 95% confidence interval describes the long-run performance of the interval construction method.

Quick revision · 6

- Probability sampling → selection based on probability laws.
- Simple random sampling → equal inclusion probabilities; successive draws are independent only with replacement.
- SRSWR → selected unit is returned before the next selection.
- SRSWOR → selected unit is not returned.
- SRS requires a complete sampling frame.

Quick revision · 7

- Stratified sampling → population divided into non-overlapping strata and samples selected independently from each stratum.
- Stratification generally improves precision when strata are internally homogeneous.
- Systematic sampling → select a random start and then every kth unit.
- Systematic sampling interval → $k = \frac{N}{n}$.
- Cluster sampling → select clusters randomly and enumerate all units within selected clusters.

Quick revision · 8

- Cluster sampling is useful when individual-unit frames are unavailable and the population is geographically dispersed.
- Probability sampling permits probability-based inference.
- Non-probability sampling → selection based on convenience, judgement, quota, or other non-random criteria.
- Convenience sampling → select easily accessible units.
- Consecutive sampling → collect every accessible eligible unit sequentially until the required number or study period is reached.

Quick revision · 9

- Quota sampling → fill predetermined subgroup quotas using non-random selection.
- Purposive sampling → researcher deliberately selects informative or suitable units.
- Snowball sampling → existing participants refer additional participants.
- Non-probability sampling can be useful when a sampling frame is unavailable or the population is difficult to reach.
- Non-probability sampling generally does not allow sampling error to be measured using probability theory.

Quick revision · 10

- Kiaer → Representative Method, 1895.
- Bowley → introduced random sampling and probability-based inference into survey work.
- P.C. Mahalanobis → pioneered large-scale sample surveys in India.
- Good survey design, representative sampling, trained data collectors, and careful execution are essential for reliable results.

Important formulae · 1

Sampling fraction:

$$f = \frac{n}{N}$$

{#eq-sampfrac}

Systematic sampling interval:

$$k = \frac{N}{n}$$

{#eq-sampinterval}

Confidence level:

$$\text{Confidence level} = 100(1-\alpha)\%$$

{#eq-conflevel}

Important formulae · 2

Expected value of an estimator:

$$E(t)$$

Unbiased estimator:

$$E(t) = \theta$$

{#eq-unbiased}

Expected value of the sample mean:

$$E(\bar{X}) = \mu$$

Self-check from the chapter

1. A numerical constant describing a population is termed a _____. **[Parameter]**
2. A numerical measure calculated from sample observations is termed a _____. **[Statistic]**
3. Complete enumeration of a population is called a _____. **[Census]**
4. The list of sampling units used for sample selection is called the sampling _____. **[Frame]**